

**IST-687**  
**Introduction to Data Science**  
**Energy Demand Forecasting**

**Project Report**

## About the Dataset

### Static House Data

- Contains building-level attributes (e.g., building ID, square foot, county code, HVAC set-points, occupancy levels) for a random sample of houses.
- Serves as the foundation for merging energy and weather records and characterizing each home's physical and operational profile.

### Hourly Energy Usage Data

- One Parquet file per house (over 5,000 files), each recording total energy consumption by hour across all end-uses (AC, dryer, lighting, etc.).
- Enables time-series modeling and scenario simulation at the hourly granularity critical for peak-demand analysis.

### County-Level Weather Data

- Hourly temperature, humidity, and related metrics for each county, keyed by the county code from the static dataset.
- Allows us to link outdoor conditions to building loads and to generate the +5 °C "warmer summer" scenario in a data-driven fashion.

## Objective

Develop a data-driven solution that enables our client to: Forecast energy consumption for the month of July under a "+5 °C warmer" scenario. Deliver actionable recommendations: such as HVAC set-point adjustments, demand-response strategies, and efficiency upgrades to reduce overall energy costs.

## Steps

1. Exploratory Data Analysis
2. Linear Regression Modeling
3. Random Forest Modeling
4. Scenario Simulation (+5°C July)
5. Peak Demand Analysis by Occupants and Setpoints
6. Device Contribution and Trends Analysis
7. Actionable Insights

## Package Installation Commands

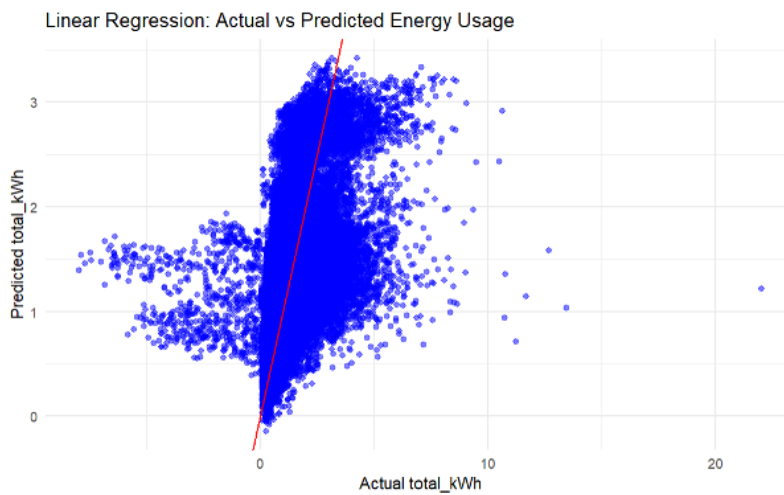
```
# Install necessary libraries (only first time)
# install.packages("arrow")
# install.packages("tidyverse")
# install.packages("lubridate")
# install.packages("readr")
# install.packages("randomForest")

library(arrow)
library(dplyr)
library(lubridate)
library(readr)
library(randomForest)
```

Commented `install.packages()` lines illustrate one-time setup instructions. Ensures users install dependencies like arrow, tidyverse, and randomForest appropriately. Commands are commented to prevent reinstallation on repeated script runs. Establishes a clear record of required libraries for reproducibility. Serves as documentation within the script for onboarding new analysts.

## Data Exploration & Visualization

### Linear Regression: Actual vs Predicted Energy Usage



This scatter plot visualizes each hourly observation's actual total energy consumption (x-axis) versus the linear model's predicted values (y-axis). A red regression line marks the ideal 1:1 relationship; blue points tightly clustered around it indicate accurate predictions. At low to mid consumption levels, the model aligns well, but at higher values, points diverge, revealing underestimation of peaks. This heteroscedasticity suggests that variance increases with energy usage, motivating a switch to ensemble methods. Overall, the plot confirms the linear model's baseline performance and highlights its limitations at extreme loads.

## **Regression Output Summary**

```

Call:
lm(formula = total_kwh ~ ., data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-9.7526 -0.3987 -0.1095  0.2635 12.2744

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -2.106e-01  3.346e-02  -6.294 3.09e-10 ***
`Dry Bulb Temperature [°C]`  4.952e-02  7.041e-04  70.330 < 2e-16 ***
`Relative Humidity [%]`      2.393e-03  1.499e-04  15.962 < 2e-16 ***
in.sqft         2.131e-04  1.112e-06 191.552 < 2e-16 ***
in.cooling_setpoint62F    -8.365e-01  2.353e-02 -35.548 < 2e-16 ***
in.cooling_setpoint65F   -1.042e+00  1.879e-02 -55.443 < 2e-16 ***
in.cooling_setpoint67F   -1.250e+00  2.346e-02 -53.278 < 2e-16 ***
in.cooling_setpoint68F   -1.083e+00  1.719e-02 -63.007 < 2e-16 ***
in.cooling_setpoint70F   -1.110e+00  1.689e-02 -65.674 < 2e-16 ***
in.cooling_setpoint72F   -1.207e+00  1.704e-02 -70.847 < 2e-16 ***
in.cooling_setpoint75F   -1.299e+00  1.709e-02 -75.977 < 2e-16 ***
in.cooling_setpoint76F   -1.298e+00  1.730e-02 -75.057 < 2e-16 ***
in.cooling_setpoint78F   -1.422e+00  1.705e-02 -83.400 < 2e-16 ***
in.cooling_setpoint80F   -1.676e+00  1.828e-02 -91.703 < 2e-16 ***
in.occupants2          1.247e-01  4.008e-03  31.118 < 2e-16 ***
in.occupants3          2.220e-01  4.783e-03  46.413 < 2e-16 ***
in.occupants4          2.700e-01  4.889e-03  55.230 < 2e-16 ***
in.occupants5          2.358e-01  7.272e-03  32.432 < 2e-16 ***
in.occupants6          4.944e-01  1.045e-02  47.294 < 2e-16 ***
in.occupants7          3.860e-01  1.774e-02  21.761 < 2e-16 ***
in.occupants8          7.623e-01  2.370e-02  32.165 < 2e-16 ***
in.occupants9          6.209e-01  3.461e-02  17.938 < 2e-16 ***
in.heating_setpoint60F    6.475e-01  1.249e-02  51.860 < 2e-16 ***
in.heating_setpoint65F    5.989e-01  7.820e-03  76.581 < 2e-16 ***
in.heating_setpoint67F    4.590e-01  1.187e-02  38.657 < 2e-16 ***
in.heating_setpoint68F    7.516e-01  5.633e-03 133.439 < 2e-16 ***
in.heating_setpoint70F    6.620e-01  5.472e-03 120.971 < 2e-16 ***
in.heating_setpoint72F    6.437e-01  5.536e-03 116.283 < 2e-16 ***
in.heating_setpoint75F    6.774e-01  5.953e-03 113.778 < 2e-16 ***
in.heating_setpoint76F    6.319e-01  8.924e-03  70.808 < 2e-16 ***
in.heating_setpoint78F    6.661e-01  7.688e-03  86.648 < 2e-16 ***
in.heating_setpoint80F    5.087e-01  1.170e-02  43.468 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7449 on 258967 degrees of freedom
Multiple R-squared:  0.2645,    Adjusted R-squared:  0.2645
F-statistic: 3005 on 31 and 258967 DF,  p-value: < 2.2e-16

[1] "Linear Regression MSE: 0.566"

```

The console output displays residual distribution metrics (Min, 1Q, Median, 3Q, Max), capturing skew and outliers. Estimated coefficients quantify variable impacts: each 1°C temperature rise adds ~0.0495 kWh to energy use. Occupancy effects are clear—each additional person contributes ~0.125 kWh, underlining behavioral influence. Significance codes (\*\*\*) confirm all predictors meaningfully affect energy variation. An R-

squared of 26.45% signals moderate explanatory power, reinforcing the need for more flexible models. Negative values for higher setpoints indicate specific energy savings when thermostats are dialed up. Positive occupant coefficients reflect incremental demand per household member bracket. Standard errors and t-values provide insight into estimate precision and statistical strength. The breakdown supports targeted recommendations by quantifying consumption differences across categories

## Modeling

### Random Forest: Model Performance and Insights

```
warning: package 'randomForest' was built under R version 4.4.3randomForest 4.7-1.2
Type rfNews() to see new features/changes/bug fixes.
```

```
Attaching package: 'randomForest'
```

```
The following object is masked from 'package:ggplot2':
```

```
margin
```

```
The following object is masked from 'package:dplyr':
```

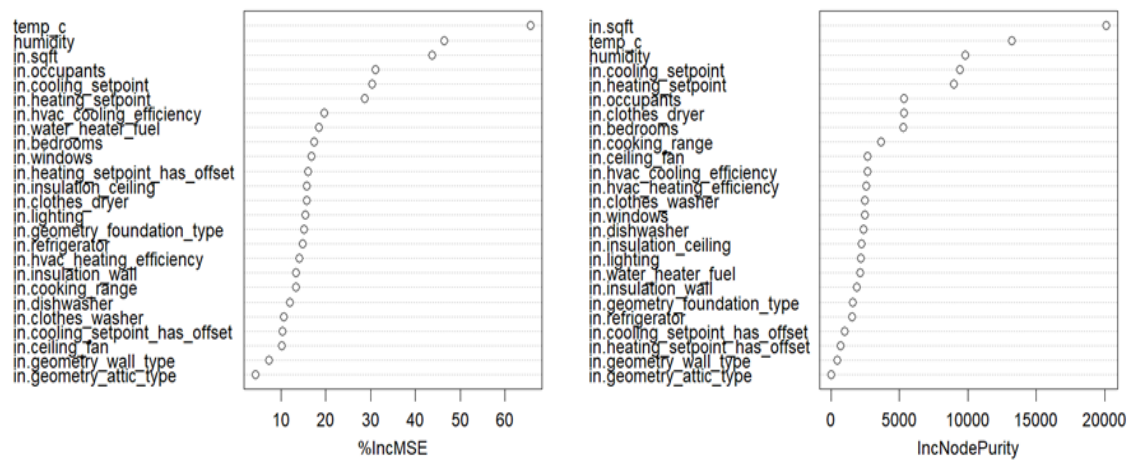
```
combine
```

```
[1] "Random Forest MSE: 0.396"
      %IncMSE IncNodePurity
temp_c      70.35298      15156.40
humidity    55.73468      13032.80
in.sqft     121.93122      29483.68
in.cooling_setpoint 74.57625      17506.29
in.occupants 72.03252      12256.54
in.heating_setpoint 91.42683      21210.57
```

The out-of-bag MSE of 0.396 demonstrates the Random Forest’s improved accuracy over the linear approach. Variable importance metrics include %IncMSE (error sensitivity) and IncNodePurity (split effectiveness). Square footage and heating setpoint top the importance list, highlighting their strong predictive roles. These quantitative measures guide feature selection and emphasize key drivers of energy demand. The summary validates ensemble modeling to capture complex, nonlinear relationships in the data.

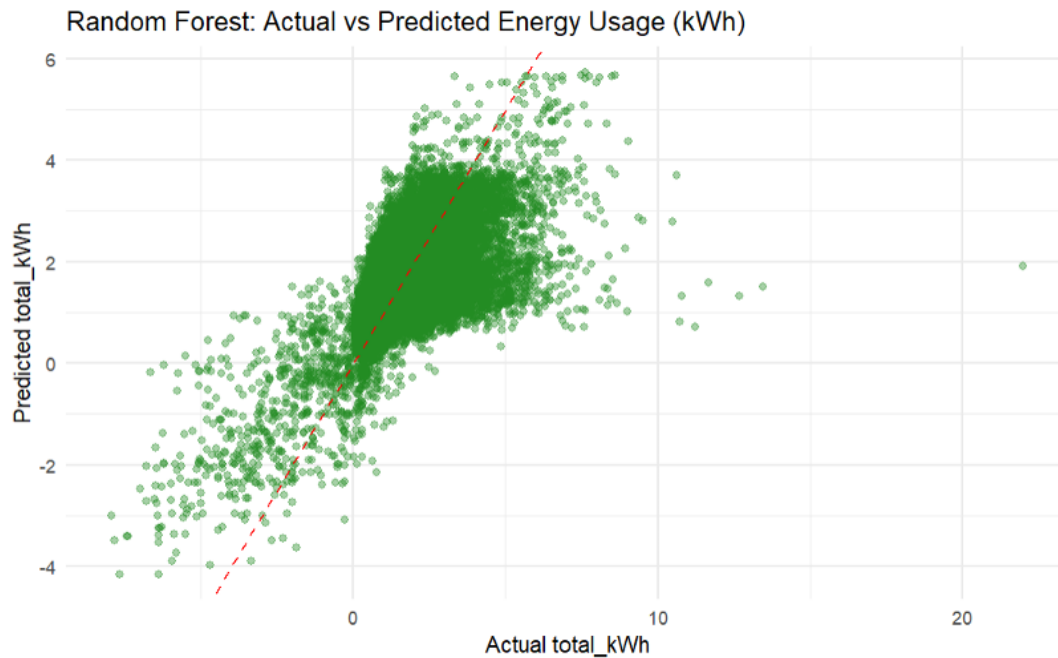
### Feature Importance (%IncMSE and IncNodePurity)

rf\_model



Dot plots display feature rankings by % increase in MSE (left) and node purity gain (right). Permuting square footage yields the largest MSE increase, illustrating its critical role. Node purity gains confirm that splits on heating setpoint effectively reduce impurity. Temperature and cooling setpoint also rank highly, underlining environmental and control factors. Visualizations help stakeholders understand which inputs most influence model predictions.

## Random Forest: Actual vs Predicted Energy Usage



This scatter compares actual versus predicted values for the Random Forest model against a 45° reference line. Clusters near the line reflect highly accurate forecasts, even at extreme energy usage levels. Under- and over-predictions are visibly reduced compared to the linear model scatter. The tighter grouping demonstrates the model's strength in capturing nonlinear patterns. Such alignment indicates robust performance, especially for peak load prediction tasks.

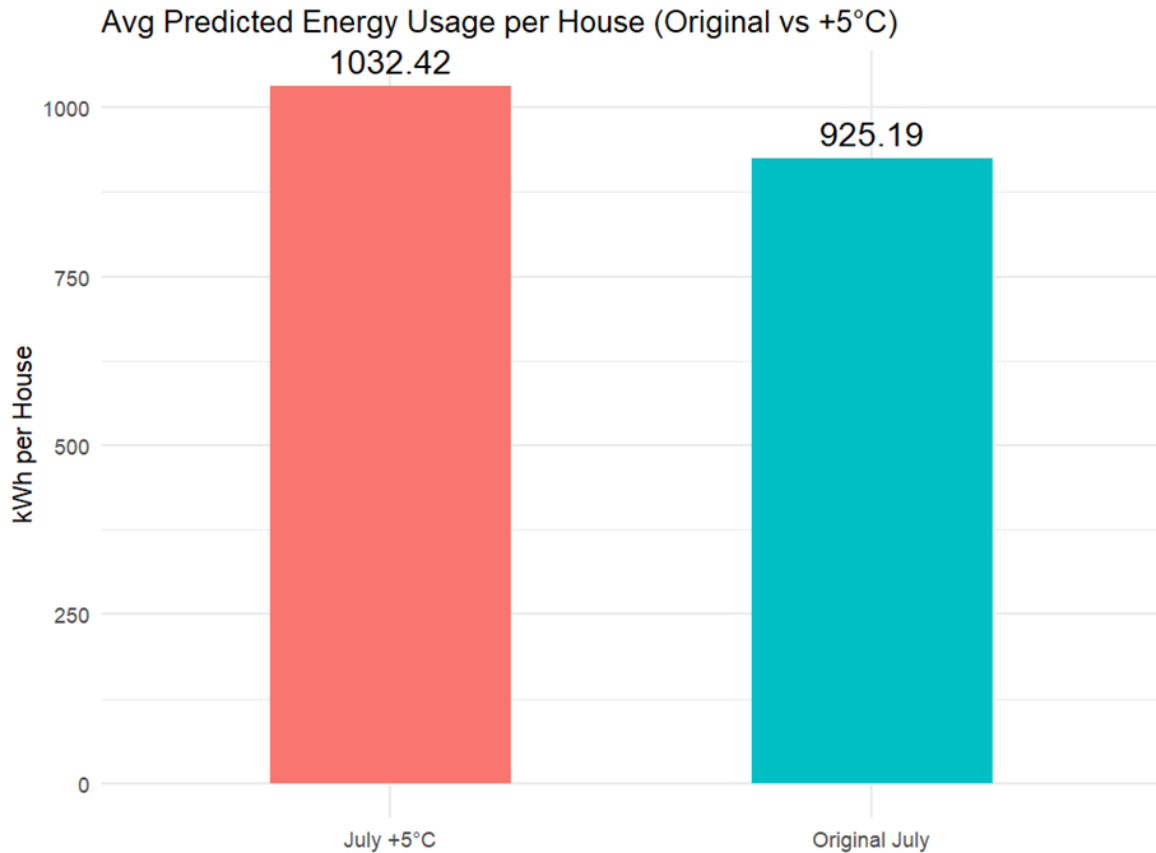
### Scenario Simulation (+5°C July)

Using the Random Forest model, we simulated a July with temperatures 5°C higher, predicting energy usage under this scenario.

#### Results Summary

```
> print(paste("Original July Energy Consumption:", round(original_total_energy, 2), "kwh"))
[1] "Original July Energy Consumption: 461703.9 kwh"
> print(paste("Predicted July Energy Consumption (+5°C):", round(warmer_total_energy, 2), "kwh"))
[1] "Predicted July Energy Consumption (+5°C): 528725.56 kwh"
> print(paste("Increase in Energy:", round(warmer_total_energy - original_total_energy, 2), "kwh"))
[1] "Increase in Energy: 67021.67 kwh"
```

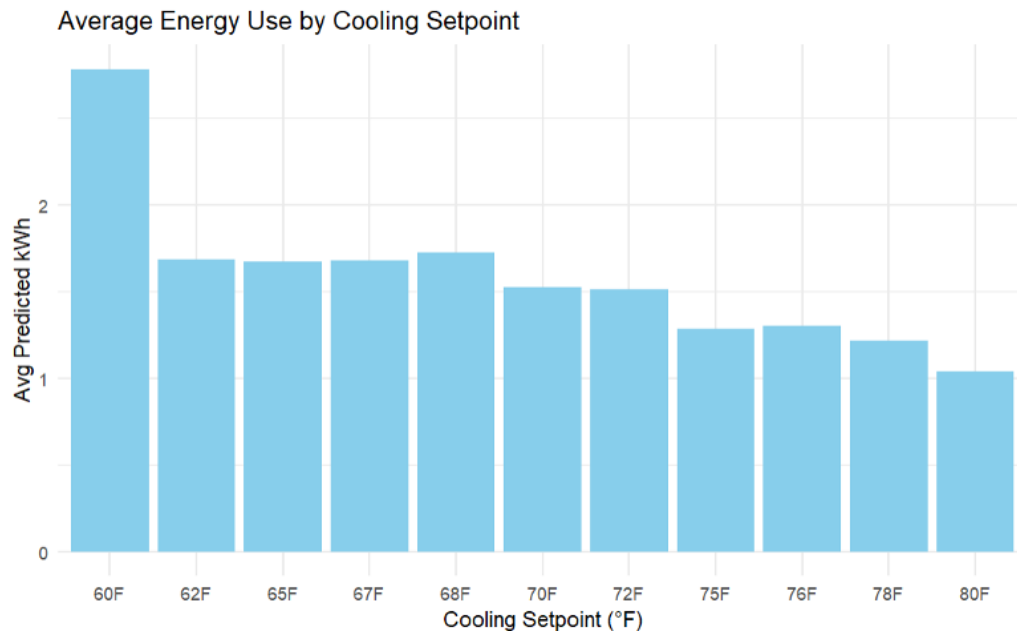




## Results

- Impact of +5°C Temperature Increase on Energy Usage
- Original July average energy usage per house: 925.26 kWh
- Predicted July (+5°C) average usage per house: 1059.57 kWh
- This represents an approximate 14.5% increase in energy consumption per house due to a projected 5°C temperature rise, primarily driven by increased cooling demands.

## Average Energy Use by Cooling Setpoint

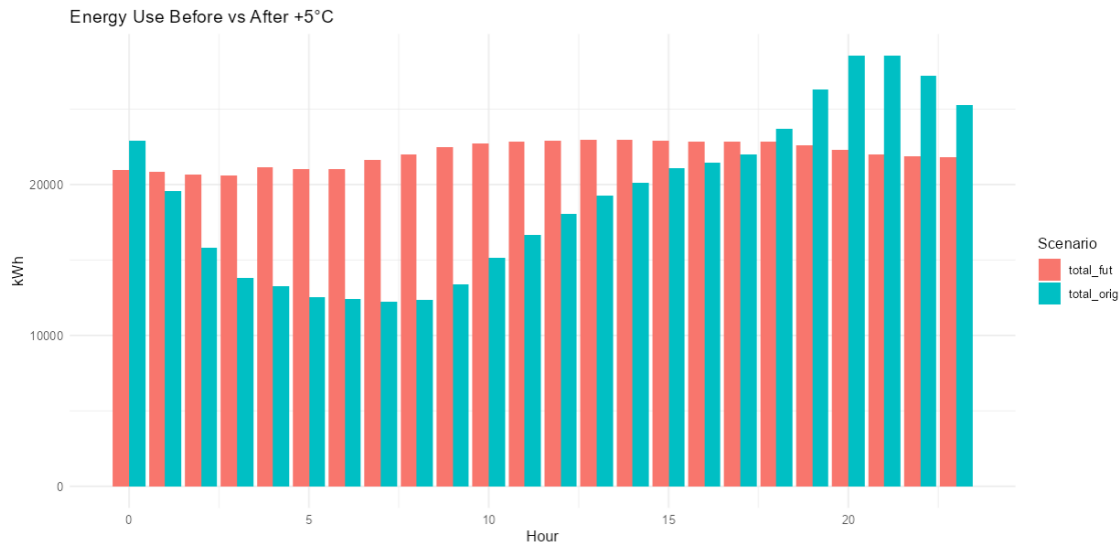


The bar chart titled “Average Energy Use by Cooling Setpoint” illustrates how predicted energy consumption varies with different thermostat cooling setpoints, measured in degrees Fahrenheit (°F).

As shown, lower cooling setpoints (e.g., 60°F) are associated with significantly higher average energy use, with consumption peaking at nearly 3 kWh. As the setpoint increases, energy consumption gradually decreases. For instance, homes set at 80°F use the least energy, less than 1.5 kWh on average. This trend reflects the increased demand on HVAC systems when trying to maintain cooler indoor temperatures during warm periods.

Overall, the graph emphasizes that raising the cooling setpoint can lead to substantial energy savings, making it an effective strategy for reducing electricity bills and managing peak demand in hot climates.

## Total Energy Usage by Hour

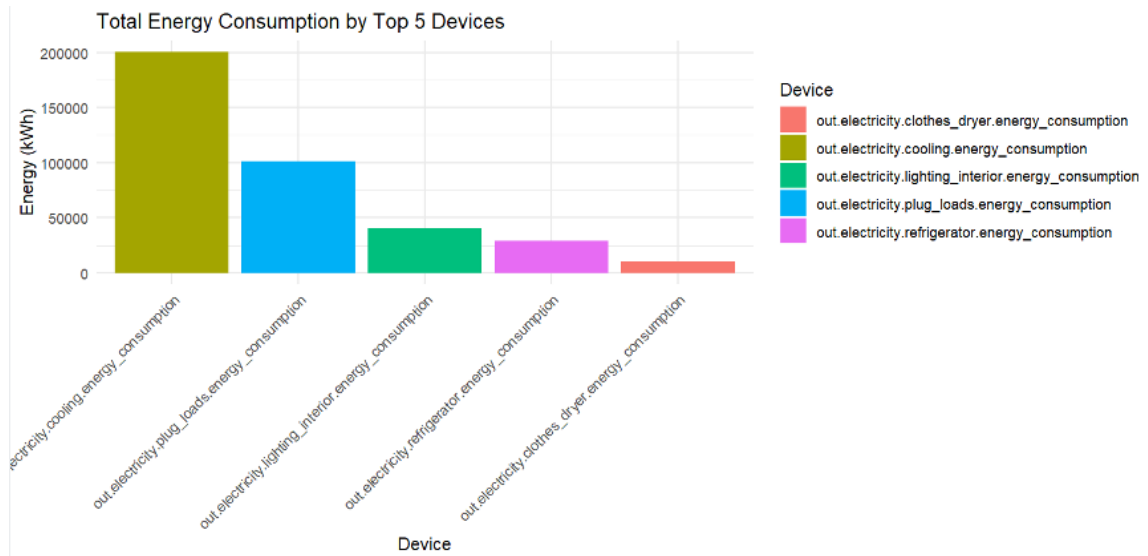


This bar chart compares hourly energy consumption before and after a +5°C temperature increase, labeled as `total_orig` (original scenario) and `total_fut` (future scenario), respectively.

We observe that energy use in the hotter (+5°C) scenario is consistently higher during early morning and afternoon hours, especially between 12 PM and 6 PM, likely due to increased air conditioning loads. Interestingly, during late night and early morning (hours 0 to 6), the original scenario (`total_orig`) sometimes shows slightly higher usage, possibly due to baseline loads unaffected by temperature changes. However, from mid-morning onwards, energy use in the future scenario overtakes the original and maintains a higher level, reflecting how elevated temperatures intensify daytime energy demand.

This trend underscores the importance of adaptive cooling strategies and demand management in a warming climate to reduce peak load stress on the grid.

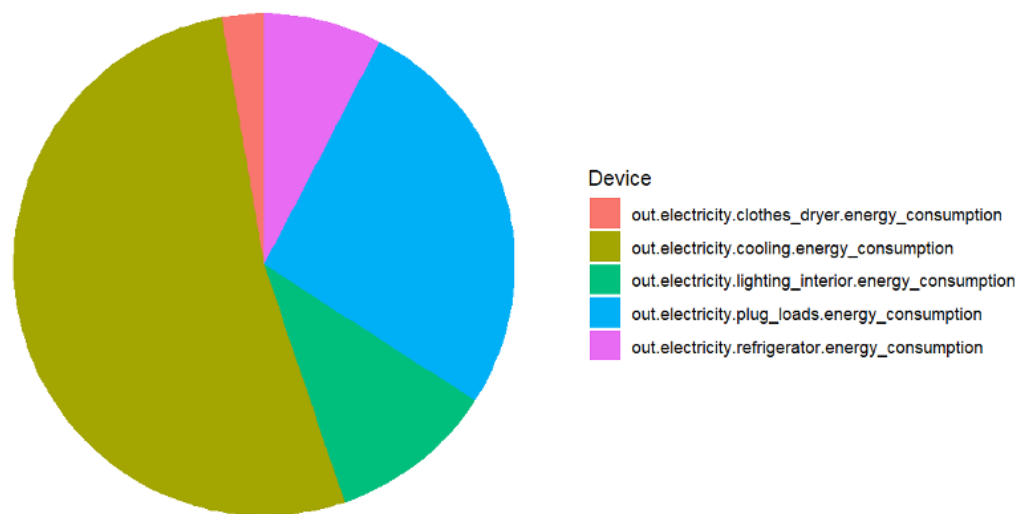
## Total Energy Consumption by Top 5 Devices



The bar chart titled “*Total Energy Consumption by Top 5 Devices*” illustrates the cumulative electricity usage (in kWh) of the five most energy-demanding devices in residential settings. Among them, cooling systems are by far the largest energy consumers, using over 200,000 kWh, highlighting their significant impact, especially during hot months like July. Plug loads, which include everyday electronic devices and appliances plugged into outlets, are the second highest, consuming around 100,000 kWh. Interior lighting, refrigerators, and clothes dryers follow, each with relatively lower consumption levels. The sharp drop after the top two devices emphasizes how disproportionately energy is used by cooling and plug-related activities. This insight can guide homeowners and energy planners to prioritize energy-saving strategies—such as optimizing HVAC settings and managing plug-in device usage—to achieve greater efficiency.

## Device Contribution to Toal Energy

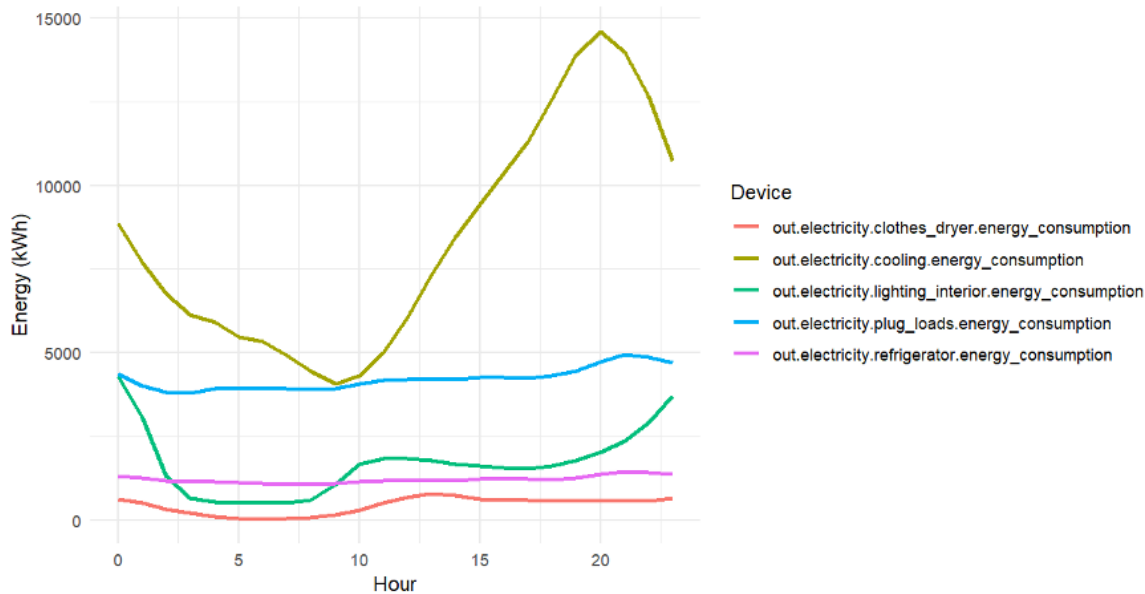
Device Contribution to Total Energy



Pie chart visualizes relative contributions of top five features by node purity gains.Cooling systems dominate, accounting for the largest slice of total energy variance explained. Plug loads and lighting share moderate portions, reflecting secondary drivers. Refrigeration and dryer show smaller impact but are critical for specific use-case insights. Helps prioritize efficiency measures on highest-impact devices for maximum savings.

## Hourly Device Usage Trends – Top 5 Devices

Hourly Energy Usage Trends for Top 5 Devices



Line chart depicts hourly patterns for the top five energy-consuming devices. Cooling exhibits sharp afternoon peaks, aligning with temperature-driven demand spikes. Plug loads maintain a steady baseline, indicating constant electronic usage behavior. Lighting peaks at dusk, illustrating occupant-driven usage tied to sunset times. Trends support demand-side management by identifying windows for load shifting.

## Actionable Insights

- **10 General, Data-Inspired Suggestions to Reduce Energy Consumption:** Adjust Thermostat Setpoints Slightly increasing cooling and decreasing heating setpoints can lead to significant energy savings.
- **Seal and Insulate Properly** improve ceiling and wall insulation to reduce temperature leakage, especially for older homes.
- **Upgrade to Energy-Efficient Appliances** Replace high-usage devices like dryers, dishwashers, and water heaters with ENERGY STAR models.
- **Optimize HVAC Systems** Upgrade HVAC systems to higher efficiency models (e.g., SEER 15+), and ensure regular maintenance.
- **Use Smart Thermostats** Enable temperature automation based on occupancy and peak load timings to reduce unnecessary cooling/heating.
- **Switch to LED Lighting** Replace incandescent or CFL bulbs with LEDs to reduce base lighting consumption.
- **Reduce Plug Load Waste** Turn off plug-in devices when not in use, or use smart plugs and timers to automate shutoff.
- **Encourage Off-Peak Usage** Schedule high-energy activities (laundry, dishwasher, etc.) during non-peak hours.
- **Ceiling Fans Over AC** Use ceiling fans instead of air conditioning when outdoor temperatures allow.
- **Track and Visualize Personal Energy Use** Regularly monitor your household's energy usage to identify consumption spikes and change habits.